

Multivariate distribution models with prescribed marginals and covariances

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Two multivariate distribution models consistent with prescribed marginal distributions and covariances are presented. The models are applicable to arbitrary number of random variables and are particularly suited for engineering applications. Conditions for validity of each model and applicable ranges of correlation coefficients between the variables are determined. Formulae are developed which facilitate evaluation of the model parameters in terms of the prescribed marginals and covariances. Potential uses of the two models in engineering are discussed.

INTRODUCTION

Multivariate distribution models are frequently needed in engineering to describe dependent random quantities. Because of the nature of engineering problems and the inadequacy of statistical data, the available information on the dependence is often limited to the knowledge of covariances between random variables. On the other hand, in many cases marginal distributions of variables can be prescribed based on relatively limited data or on physical or mathematical grounds, e.g., the central limit theorem or asymptotic theorems of extreme values. In such situations, there is a need for multivariate distribution models which are consistent with the set of known marginals and covariances. Such models, of course, can be further tested with standard statistical techniques against any available data to verify their appropriateness.

Beginning with Frechet's pioneering work in 1951¹, there has been a growing number of joint distribution models with prescribed marginals (see Johnson and Kotz²). Most existing models, however, are restricted to the bivariate case and/or can only describe small correlation between the variables. In this paper, two multivariate distribution models are presented which are based on earlier works of Morgenstern³ and Nataf⁴. Besides being applicable to an arbitrary number of random variables, these models are easily evaluated in terms of the known marginals and correlation coefficients of the variables. The conditions for validity of the two models are examined and the range of correlation coefficients that can be described are determined. Potential uses of these models in engineering, particularly in the assessment of structural reliability, are discussed.

Of special interest in this paper are marginal distributions which are described by at most two parameters. Among these, two groups of distributions are identified: Distributions which are reducible to a standard form through a linear transformation of the variable, denoted herein as Group 1 distributions, and distributions which are not reducible to a standard form,

denoted herein as Group 2 distributions. Several examples for these two groups of distributions are listed in Table 1.

THE MORGENSTERN MODEL

For two variables X_1 and X_2 with marginal cumulative distribution functions (CDF's) $F_{X_1}(x_1)$ and $F_{X_2}(x_2)$, Morgenstern's bivariate CDF is³

$$F_{X_1, X_2}(x_1, x_2) = F_{X_1}(x_1)F_{X_2}(x_2)\{1 + \alpha[1 - F_{X_1}(x_1)][1 - F_{X_2}(x_2)]\} \quad (1)$$

The corresponding joint probability density function (PDF) is

$$f_{X_1, X_2}(x_1, x_2) = \frac{\partial^2 F_{X_1, X_2}(x_1, x_2)}{\partial x_1 \partial x_2} = f_{X_1}(x_1)f_{X_2}(x_2)\{1 + \alpha[1 - 2F_{X_1}(x_1)][1 - 2F_{X_2}(x_2)]\} \quad (2)$$

where $f_{X_i}(x_i) = \partial F_{X_i}(x_i)/\partial x_i$ are the marginal PDF's. This model is valid if $f_{X_1, X_2}(x_1, x_2) \geq 0$, which leads to the requirement $|\alpha| \leq 1$. The parameter α is related to the correlation coefficient, ρ_{12} , of X_1 and X_2 through

$$\rho_{12} = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \left(\frac{x_1 - \mu_1}{\sigma_1} \right) \left(\frac{x_2 - \mu_2}{\sigma_2} \right) f_{X_1, X_2}(x_1, x_2) dx_1 dx_2 = 4\alpha Q_1 Q_2 \quad (3)$$

where μ_i and σ_i are the mean and standard deviation of X_i , respectively, and

$$Q_i = \int_{-\infty}^{\infty} \left(\frac{x_i - \mu_i}{\sigma_i} \right) f_{X_i}(x_i) F_{X_i}(x_i) dx_i \quad (4)$$

This result is obtained by substituting from equation (2) into equation (3) and observing that all integrals of the

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Table 1. Selected two-parameter distributions

Group	Name	Symbol	CDF	Standard CDF
1	Normal	N	$\Phi\left(\frac{x-\mu}{\sigma}\right)$	$\Phi(y)$
	Uniform	U	$\frac{x-a}{b-a}, \quad a \leq x \leq b$	$y, 0 \leq y \leq 1$
	Shifted exponential	SE	$1 - \exp[-\lambda(x-x_0)], \quad x_0 \leq x$	$1 - \exp(-y), \quad 0 \leq y$
	Shifted Rayleigh	SR	$1 - \exp\left[-\frac{1}{2}\left(\frac{x-x_0}{\alpha}\right)^2\right], \quad x_0 \leq x$	$1 - \exp\left[-\frac{y^2}{2}\right], \quad 0 \leq y$
	Type-I largest value	T1L	$\exp\{-\exp[-\alpha(x-u)]\}$	$\exp[-\exp(-y)]$
	Type-I smallest value	T1S	$1 - \exp\{-\exp[\alpha(x-u)]\}$	$1 - \exp[-\exp(y)]$
2	Lognormal	LN	$\Phi\left[\frac{\ln x - \lambda}{\zeta}\right], \quad 0 < x$	
	Gamma	GM	$\frac{\Gamma(k, \lambda x)}{\Gamma(k)}, \quad 0 \leq x$	
	Type-II largest value	T2L	$\exp\left[-\left(\frac{u}{x}\right)^k\right], \quad 0 < x$	
	Type-III smallest value	T3S	$1 - \exp\left[-\left(\frac{x-e}{u-e}\right)^k\right], \quad e \leq x$	

Note: $\Phi(\cdot)$ is the standard normal probability; $\Gamma(k, u) = \int_0^u e^{-u} u^{k-1} du$ and $\Gamma(k) = \Gamma(k, \infty)$ are the incomplete and complete gamma functions, respectively

form $\int_{-\infty}^{\infty} [(x_i - \mu_i)/\sigma_i] f_X(x_i) dx_i$ in equation (3) are identically zero. Thus,

$$\alpha = \frac{\rho_{12}}{4Q_1Q_2} \quad (5)$$

Expressions for Q_i for the distributions in Table 1 are listed in Table 2. It is interesting to note that for Group 1 distributions Q_i are constants, whereas for Group 2 distributions Q_i are functions of the shape parameter of the distribution.

It is shown in Appendix A that in general $0 < Q_i < 0.5$. However, for most distributions $Q_i < 0.3$ (see Table 2). It follows from equation (5) that, since $|\alpha| \leq 1$, the range of correlation coefficients that can be described by Morgenstern's model is rather limited. Table 3 lists maximum permitted $|\rho_{12}|$ for the selected distributions. When either marginal distribution belongs to Group 2, the maximum permitted correlation depends on the shape parameter. For such cases, the upper bounds of the maximum permitted $|\rho_{12}|$ are indicated in Table 3. The model is seen to be applicable only to variables with rather low correlation, i.e., $|\rho_{12}|$ less than around 0.3.

The following generalization of Morgenstern's model for n variables $\mathbf{X} = (X_1, \dots, X_n)$ has been suggested by Kotz⁵,

$$F_{\mathbf{X}}(\mathbf{x}) = \prod_i F_{X_i}(x_i) \left\{ 1 + \sum_{i < j} \alpha_{ij} [1 - F_{X_i}(x_i)] [1 - F_{X_j}(x_j)] + \sum_{i < j < k} \alpha_{ijk} [1 - F_{X_i}(x_i)] [1 - F_{X_j}(x_j)] [1 - F_{X_k}(x_k)] + \dots \right\} \quad (6)$$

The probability density function is a generalization of equation (2), i.e.,

$$f_{\mathbf{X}}(\mathbf{x}) = \prod_i f_{X_i}(x_i) \left\{ 1 + \sum_{i < j} \alpha_{ij} [1 - 2F_{X_i}(x_i)] [1 - 2F_{X_j}(x_j)] + \sum_{i < j < k} \alpha_{ijk} [1 - 2F_{X_i}(x_i)] [1 - 2F_{X_j}(x_j)] \times [1 - 2F_{X_k}(x_k)] + \dots \right\} \quad (7)$$

Table 2. $Q = \int_{-\infty}^{\infty} \left(\frac{x-\mu}{\sigma}\right) f_X(x) F_X(x) dx$ for selected distributions

Distribution	Q
Normal	$\frac{1}{2\sqrt{\pi}} = 0.282$
Uniform	$\frac{\sqrt{3}}{6} = 0.289$
Shifted exponential	$\frac{1}{4} = 0.25$
Shifted Rayleigh	$\frac{\sqrt{2\pi} - \sqrt{\pi}}{\sqrt{32 - 8\pi}} = 0.280$
Type-I largest value	$\frac{\sqrt{6 \ln 6}}{2\pi} = 0.270$
Type-I smallest value	$\frac{\sqrt{6 \ln 2}}{2\pi} = 0.270$
Lognormal	$\frac{2\Phi(\zeta/\sqrt{2}) - 1}{2\sqrt{\exp(\zeta^2) - 1}} < 0.282$
Gamma	$\frac{\Gamma(2k)}{4^k \sqrt{k} \Gamma^2(k)} < 0.282$
Type-II largest value	$\frac{(2^{1/k} - 1)\Gamma(1 - 1/k)}{2\sqrt{\Gamma(1 - 2/k) - \Gamma^2(1 - 1/k)}} < 0.270$
Type-III smallest value	$\frac{(1 - 2^{-1/k})\Gamma(1 + 1/k)}{2\sqrt{\Gamma(1 + 2/k) - \Gamma^2(1 + 1/k)}} < 0.270$

Note: $\Phi(\cdot)$ is the standard normal probability and $\Gamma(\cdot)$ is the gamma function

Here, $|\alpha_{ij}| \leq 1$ are only sufficient conditions to satisfy the nonnegativeness of the bivariate PDF of each pair X_i and X_j . Sufficient conditions on all parameters to ensure the validity of the model are obtained by setting $f_{\mathbf{X}}(\mathbf{x}) \geq 0$. For each i and j , the bivariate distribution of X_i and X_j from

Table 3. Maximum permitted $|\rho_{12}|$ for Morgenstern Model

Marginal distribution	N	U	SE	SR	T1L	T1S	LN	GM	T2L	T3S
N	0.318									
U	0.326	0.333								
SE	0.282	0.289	0.250							
SR	0.316	0.324	0.280	0.314						
T1L	0.305	0.312	0.270	0.303	0.292					
T1S	0.305	0.312	0.270	0.303	0.292	0.292				
LN	<0.318	<0.326	<0.282	<0.316	<0.305	<0.305	<0.318			
GM	<0.318	<0.326	<0.282	<0.316	<0.305	<0.305	<0.318	<0.318		
T2L	<0.305	<0.312	<0.270	<0.303	<0.292	<0.292	<0.305	<0.305	<0.292	
T3S	<0.305	<0.312	<0.270	<0.303	<0.292	<0.292	<0.305	<0.305	<0.292	<0.292

the preceding equation is of the same form as that in equation (2) with α_{ij} replacing α . It follows that α_{ij} are obtained from equation (5) with the subscripts 1 and 2 replaced by i and j , respectively. Furthermore, it is easily verified that α_{ijk} , etc., are obtained from generalization of equation (5) in the form

$$\alpha_{12\dots k} = \frac{\rho_{12\dots k}}{(-2)^k Q_1 Q_2 \dots Q_k} \quad (8)$$

where $\rho_{12\dots k}$ is the k -dimensional, normalized, joint central moment

$$\rho_{12\dots k} = E \left[\left(\frac{X_1 - \mu_1}{\sigma_1} \right) \left(\frac{X_2 - \mu_2}{\sigma_2} \right) \dots \left(\frac{X_k - \mu_k}{\sigma_k} \right) \right] \quad (9)$$

These higher-order moments, however, are seldom available in practice. It is appropriate then to truncate the series in equation (6) at a level beyond which the joint moments are not known.

For each set of α_{ij} , etc., obtained from equation (8), the joint PDF in equation (7) should be examined to verify the nonnegativeness of the PDF. Because of this condition, the range of allowable correlation coefficients in the multivariate case, in general, can be narrower than those for the bivariate case listed in Table 3. This limitation may prohibit the use of the Morgenstern model in many applications. The model presented in the following section is applicable to a wider range of correlation coefficients.

THE NATAF MODEL

Define standard normal variates $\mathbf{Z} = (Z_1, \dots, Z_n)$ obtained by marginal transformations of $\mathbf{X} = (X_1, \dots, X_n)$.

$$Z_i = \Phi^{-1}[F_{X_i}(X_i)], \quad i = 1, \dots, n \quad (10)$$

where $\Phi(\cdot)$ is the standard cumulative normal probability. Nataf's⁴ distribution for $\mathbf{X}$ is obtained by assuming that $\mathbf{Z}$ is jointly normal. Using the rules of probability transformation, the joint PDF of $\mathbf{X}$ is

$$f_{\mathbf{X}}(\mathbf{x}) = f_{X_1}(x_1) f_{X_2}(x_2) \dots f_{X_n}(x_n) \frac{\varphi_n(\mathbf{z}, \mathbf{R}')}{\varphi(z_1) \varphi(z_2) \dots \varphi(z_n)} \quad (11)$$

where $z_i = \Phi^{-1}[F_{X_i}(x_i)]$, $\varphi(\cdot)$ is the standard normal PDF, and $\varphi_n(\mathbf{z}, \mathbf{R}')$ in the n -dimensional normal PDF of zero means, unit standard deviations, and correlation matrix $\mathbf{R}'$. The elements ρ'_{ij} of $\mathbf{R}'$ are defined in terms of the

correlation coefficients ρ_{ij} through the integral relation

$$\begin{aligned} \rho_{ij} &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \left(\frac{x_i - \mu_i}{\sigma_i} \right) \left(\frac{x_j - \mu_j}{\sigma_j} \right) f_{X_i}(x_i) f_{X_j}(x_j) \\ &\quad \times \frac{\varphi_2(z_i, z_j, \rho'_{ij})}{\varphi(z_i) \varphi(z_j)} dx_i dx_j \\ &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \left(\frac{x_i - \mu_i}{\sigma_i} \right) \left(\frac{x_j - \mu_j}{\sigma_j} \right) \varphi_2(z_i, z_j, \rho'_{ij}) dz_i dz_j \end{aligned} \quad (12)$$

The preceding distribution model is valid provided the mappings in equation (10) are one to one and the correlation matrix $\mathbf{R}'$ is positive definite. The first condition is satisfied when $F_{X_i}(x_i)$ are continuous and strictly increasing. The second condition is satisfied in nearly all cases of practical interest, since the correlation matrix of $\mathbf{X}$ by definition is positive definite and, as subsequently shown, the differences $|\rho_{ij} - \rho'_{ij}|$ are small.

For each pair of marginal distributions and with known ρ_{ij} , equation (12) can be iteratively solved for ρ'_{ij} . To avoid such tedious calculations, a set of empirical formulae relating ρ'_{ij} to ρ_{ij} are developed⁶. These are based on the following fundamental properties of equation (12):

Lemma 1. ρ_{ij} is a strictly increasing function of ρ'_{ij} .

Lemma 2. $\rho'_{ij} = 0$ for $\rho_{ij} = 0$.

Lemma 3. $|\rho_{ij}| \leq |\rho'_{ij}|$ where the equality stands when $\rho_{ij} = 0$ or when both marginals are normal.

Proofs for the above Lemmas are presented in Appendix B. Based on these properties, the solution of ρ'_{ij} in equation (12) is unique and can be expressed in the form

$$\rho'_{ij} = \rho_{ij} F \quad (13)$$

in which F , satisfying $F \geq 1$, is a function of ρ_{ij} and the marginal distributions of X_i and X_j . The function F has the following additional properties (see Appendix B for the proofs):

Lemma 4. F is independent of ρ_{ij} if one of the variables is normal.

Lemma 5. F is invariant to increasing linear transformations of X_i and X_j .

Lemma 6. F is independent of the parameters of Group 1 distributions.

Lemma 7. F is a function of the coefficient of variation, $\delta = \sigma/\mu$, of Group 2 distributions.

Based on Lemmas 4–7, five categories of formulae for F for two-parameter distributions are developed: (1) $F = \text{constant}$ for X_j belonging to Group 1 and X_i normal; (2) $F = F(\delta_j)$ for X_j belonging to Group 2 and X_i normal; (3) $F = F(\rho_{ij})$ for both X_i and X_j belonging to Group 1; (4) $F = F(\rho_{ij}, \delta_j)$ for X_i belonging to Group 1 and X_j belonging to Group 2; and (5) $F = F(\rho_{ij}, \delta_i, \delta_j)$ for both X_i and X_j belonging to Group 2. For the selected distributions in Table 1, the formulae are developed by least-square fitting of polynomial expressions to exact values computed by numerical integration of equation (12). The results are listed in Tables 4–8 together with maximum errors resulting from the least-square fitting. The range of coefficients of variation used in generating the formulae in Tables 5, 7, and 8 is 0.1–0.5. For values outside this range, the errors in the formulae can be larger than those listed in these tables.

As demonstrations of the formulae in Tables 4–8, plots of F for several selected pairs of marginal distributions are shown in Figs 1–3 together with the exact results obtained from numerical integration of equation (12). Figs. 1 and 2 represent typical cases, whereas Fig. 3 represents the case with poorest agreement with exact results (maximum error = 4.5%, Table 7). For most distributions F is only slightly higher than unity. However, it can be as high as 1.5 or greater when X_i and X_j are negatively correlated and

Table 4. Category 1 formulae, $F = \text{constant}$, for X_j belonging to group 1 and X_i normal

X_j	$F = \text{constant}$	Max. error
Uniform	1.023	0.0%
Shifted exponential	1.107	0.0%
Shifted Rayleigh	1.014	0.0%
Type-I largest value	1.031	0.0%
Type-I smallest value	1.031	0.0%

Table 5. Category 2 formulae, $F = F(\delta_i)$, for X_j belonging to group 2 and X_i normal

X_j	$F = F(\delta_j)$	Max. error
Lognormal	$\frac{\delta_j}{\sqrt{\ln(1 + \delta_j^2)}}$	(exact)
Gamma	$1.001 - 0.007\delta_j + 0.118\delta_j^2$	0.0%
Type-II largest value	$1.030 + 0.238\delta_j + 0.364\delta_j^2$	0.1%
Type-III smallest value	$1.031 - 0.195\delta_j + 0.328\delta_j^2$	0.1%

Note: range of coefficient of variation is $\delta_j = 0.1 - 0.5$

Table 6. Category 3 formulae, $F = F(\rho_{ij})$, for X_i and X_j both belonging to group 1

$X_i \backslash X_j$	U	SE	SR	T1L	T1S
U	$1.047 - 0.047\rho^2$				
(Max. error)	(0.0%)				
SE	$1.133 + 0.029\rho^2$	$1.229 - 0.367\rho + 0.153\rho^2$			
(Max. error)	(0.0%)	(1.5%)			
SR	$1.038 - 0.008\rho^2$	$1.123 - 0.100\rho + 0.021\rho^2$	$1.028 - 0.029\rho$		
(Max. error)	(0.0%)	(0.1%)	(0.0%)		
T1L	$1.055 + 0.015\rho^2$	$1.142 - 0.154\rho + 0.031\rho^2$	$1.046 - 0.045\rho + 0.006\rho^2$	$1.064 - 0.069\rho + 0.005\rho^2$	
(Max. error)	(0.0%)	(0.2%)	(0.0%)	(0.0%)	
T1S	$1.055 + 0.015\rho^2$	$1.142 + 0.154\rho + 0.031\rho^2$	$1.046 + 0.045\rho + 0.006\rho^2$	$1.064 + 0.069\rho + 0.005\rho^2$	$1.064 - 0.069\rho + 0.005\rho^2$
(Max. error)	(0.0%)	(0.2%)	(0.0%)	(0.0%)	(0.0%)

Note: $\rho = \rho_{ij}$

their marginals are skewed in the same direction (e.g., Fig. 2 for exponential X_i and X_j) or when they are positively correlated and their marginals are skewed in opposite directions.

In contrast to the Morgenstern model, the distribution model in equation (11) is applicable to a rather wide range of correlation coefficients. Table 9 lists the maximum permitted ranges of ρ_{ij} for Group 1 distributions. For Group 2 distributions, the permitted range is dependent on the coefficients of variation, but the results are of similar nature. For more than two variables, the actual permitted range may be smaller to satisfy the positive definiteness of $\mathbf{R}'$.

EXAMPLES

Let variables X_1 and X_2 be marginally standard normals with correlation coefficient 0.3. From equation (5) and Table 2, $\alpha = \pi\rho_{12}$. Hence, the bivariate PDF based on the Morgenstern model is

$$f_{X_1, X_2}(x_1, x_2) = \frac{1}{2\pi} \exp\left(-\frac{x_1^2 + x_2^2}{2}\right) \times \{1 + 0.3\pi[1 - 2\Phi(x_1)][1 - 2\Phi(x_2)]\} \quad (14)$$

Applying the Nataf model yields the bivariate normal distribution with $\rho'_{12} = \rho_{12} = 0.3$, i.e.,

$$f_{X_1, X_2}(x_1, x_2) = \frac{1}{2\pi\sqrt{1 - 0.09}} \exp\left(\frac{x_1^2 - 0.6x_1x_2 + x_2^2}{2(1 - 0.09)}\right) \quad (15)$$

For this special case, the PDF's based on the two models are both well behaving surfaces and are nearly coincident.

As a second example, let X_1 and X_2 have identical exponential marginals with means 1.0 and correlation coefficients 0.25. (This is the highest correlation that can be used with the Morgenstern model.) From equation (5) and Table 2, $\alpha = 1$ and the bivariate PDF based on the Morgenstern model is

$$f_{X_1, X_2}(x_1, x_2) = \exp(-x_1 - x_2) \times \{1 + [2\exp(-x_1) - 1][2\exp(-x_2) - 1]\} \quad (16)$$

For the Nataf model, the correlation coefficient

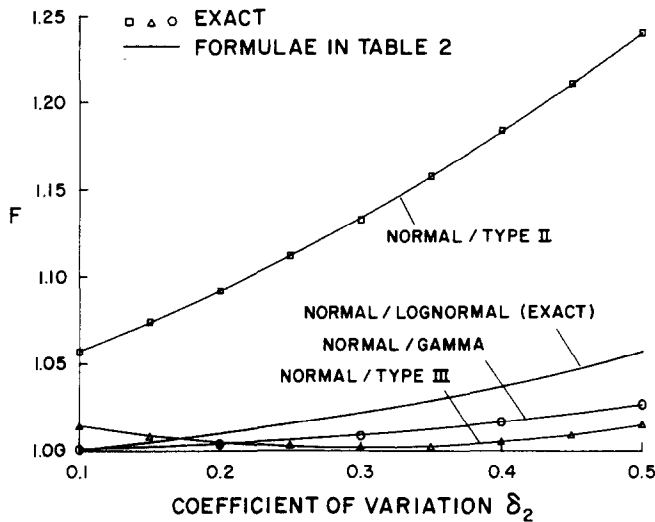
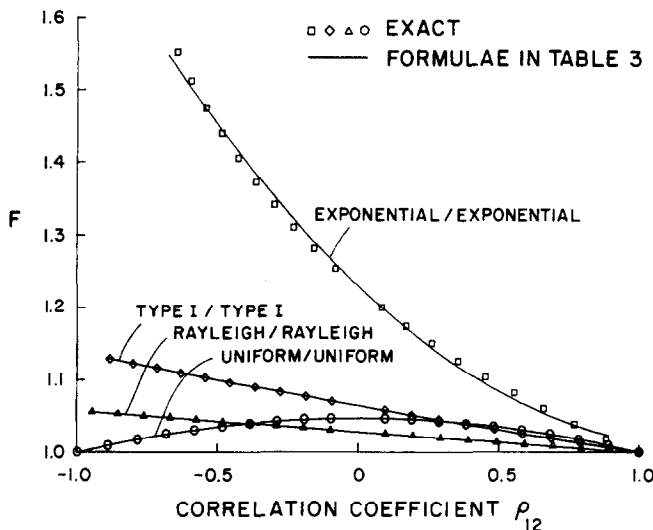
Table 7. Category 4 formulae, $F = F(\rho_{ij}, \delta_j)$, for X_i belonging to group 1 and X_j belonging to group 2

$X_j \backslash X_i$	U	SE	SE	T1L	T1S
LN	$1.019 + 0.014\delta_j$ $+ 0.010\rho^2 + 0.249\delta_j^2$ (0.7%)	$1.098 + 0.003\rho + 0.019\delta_j$ $+ 0.025\rho^2 + 0.303\delta_j^2 - 0.437\rho\delta_j$ (1.6%)	$1.011 + 0.001\rho + 0.014\delta_j$ $+ 0.004\rho^2 + 0.231\delta_j^2 - 0.130\rho\delta_j$ (0.4%)	$1.029 + 0.001\rho + 0.014\delta_j$ $+ 0.004\rho^2 + 0.233\delta_j^2 - 0.197\rho\delta_j$ (0.3%)	$1.029 - 0.001\rho + 0.014\delta_j$ $+ 0.004\rho^2 + 0.233\delta_j^2 + 0.197\rho\delta_j$ (0.3%)
GM	$1.023 - 0.007\delta_j$ $+ 0.002\rho^2 + 0.127\delta_j^2$ (0.1%)	$1.104 + 0.003\rho - 0.008\delta_j$ $+ 0.014\rho^2 + 0.173\delta_j^2 - 0.296\rho\delta_j$ (0.9%)	$1.014 + 0.001\rho - 0.007\delta_j$ $+ 0.002\rho^2 + 0.126\delta_j^2 - 0.090\rho\delta_j$ (0.9%)	$1.031 + 0.001\rho - 0.007\delta_j$ $+ 0.003\rho^2 + 0.131\delta_j^2 - 0.132\rho\delta_j$ (0.3%)	$1.031 - 0.001\rho - 0.007\delta_j$ $+ 0.003\rho^2 + 0.131\delta_j^2 + 0.132\rho\delta_j$ (0.3%)
T2L	$1.033 + 0.305\delta_j$ $+ 0.074\rho^2 + 0.405\delta_j^2$ (2.1%)	$1.109 - 0.152\rho + 0.361\delta_j$ $+ 0.130\rho^2 + 0.455\delta_j^2 - 0.728\rho\delta_j$ (4.5%)	$1.036 - 0.038\rho + 0.266\delta_j$ $+ 0.028\rho^2 + 0.383\delta_j^2 - 0.229\rho\delta_j$ (1.2%)	$1.056 - 0.060\rho + 0.263\delta_j$ $+ 0.020\rho^2 + 0.383\delta_j^2 - 0.332\rho\delta_j$ (1.0%)	$1.056 + 0.060\rho + 0.263\delta_j$ $+ 0.020\rho^2 + 0.383\delta_j^2 + 0.332\rho\delta_j$ (1.0%)
T3S	$1.061 - 0.237\delta_j$ $- 0.005\rho^2 + 0.379\delta_j^2$ (0.5%)	$1.147 + 0.145\rho - 0.271\delta_j$ $+ 0.010\rho^2 + 0.459\delta_j^2 - 0.467\rho\delta_j$ (0.4%)	$1.047 + 0.042\rho - 0.212\delta_j$ $+ 0.353\delta_j^2 - 0.136\rho\delta_j$ (0.2%)	$1.064 + 0.065\rho - 0.210\delta_j$ $+ 0.003\rho^2 + 0.356\delta_j^2 - 0.211\rho\delta_j$ (0.2%)	$1.064 - 0.065\rho - 0.210\delta_j$ $+ 0.003\rho^2 + 0.356\delta_j^2 + 0.211\rho\delta_j$ (0.2%)

Note: $\rho = \rho_{ij}$; range of coefficient of variation is $\delta_j = 0.1 - 0.5$ Table 8. Category 5 formulae, $F = F(\rho_{ij}, \delta_i, \delta_j)$, for X_i and X_j both belonging to group 2

$X_j \backslash X_i$	LN	GM	T2L	T3S
LN	$\frac{\ln(1 + \rho\delta_j)}{\rho\sqrt{\ln(1 + \delta_i^2)\ln(1 + \delta_j^2)}}$ (exact)			
(Max. error)				
GM	$1.001 + 0.033\rho + 0.004\delta_i - 0.016\delta_j$ $+ 0.002\rho^2 + 0.223\delta_i^2 + 0.130\delta_j^2$ $- 0.104\rho\delta_i + 0.029\delta_i\delta_j - 0.119\rho\delta_j$ (4.0%)	$1.002 + 0.022\rho - 0.012\delta_i + \delta_j$ $+ 0.001\rho^2 + 0.125(\delta_i^2 + \delta_j^2)$ $- 0.077\rho(\delta_i + \delta_j) + 0.014\delta_i\delta_j$ (4.0%)	$1.086 + 0.054\rho + 0.104(\delta_i + \delta_j)$ $- 0.055\rho^2 + 0.662(\delta_i^2 + \delta_j^2)$ $- 0.570\rho(\delta_i + \delta_j) + 0.203\delta_i\delta_j$ $- 0.020\rho^3 - 0.218(\delta_i^3 + \delta_j^3)$ $- 0.371\rho(\delta_i^2 + \delta_j^2) + 0.257\rho^2(\delta_i + \delta_j)$ $+ 0.141\delta_i\delta_j(\delta_i + \delta_j)$ (4.3%)	
(Max. error)				
T2L	$1.026 + 0.082\rho - 0.019\delta_i + 0.222\delta_j$ $+ 0.018\rho^2 + 0.288\delta_i^2 + 0.379\delta_j^2$ $- 0.441\rho\delta_i + 0.126\delta_i\delta_j - 0.277\rho\delta_j$	$1.029 + 0.056\rho - 0.030\delta_i + 0.225\delta_j$ $+ 0.012\rho^2 + 0.174\delta_i^2 + 0.379\delta_j^2$ $- 0.313\rho\delta_i + 0.075\delta_i\delta_j - 0.182\rho\delta_j$		
(Max. error)				
T3S	$1.031 + 0.052\rho + 0.011\delta_i - 0.210\delta_j$ $+ 0.002\rho^2 + 0.220\delta_i^2 + 0.350\delta_j^2$ $+ 0.005\rho\delta_i + 0.009\delta_i\delta_j - 0.174\rho\delta_j$ (2.4%)	$1.032 + 0.034\rho - 0.007\delta_i - 0.202\delta_j$ $+ 0.121\delta_i^2 + 0.339\delta_j^2$ $- 0.006\rho\delta_i + 0.003\delta_i\delta_j - 0.111\rho\delta_j$ (4.0%)	$1.065 + 0.146\rho + 0.241\delta_i - 0.259\delta_j$ $+ 0.013\rho^2 + 0.372\delta_i^2 + 0.435\delta_j^2$ $+ 0.005\rho\delta_i + 0.034\delta_i\delta_j - 0.481\rho\delta_j$ (3.8%)	$1.063 - 0.004\rho - 0.200\delta_i + \delta_j$ $- 0.001\rho^2 + 0.337(\delta_i^2 + \delta_j^2)$ $+ 0.007\rho(\delta_i + \delta_j) - 0.007\delta_i\delta_j$ (2.6%)

Note: $\rho = \rho_{ij}$; ranges of coefficients of variation are $\delta_i, \delta_j = 0.1 - 0.5$


Fig. 1. F for X_2 belonging to group 2 and X_1 normal

Fig. 2. Selected plots of F for X_1 and X_2 both belonging to group 1

$\rho'_{12} = 0.287$ is obtained from Table 6 and the bivariate PDF is

$$f_{X_1, X_2}(x_1, x_2) = 1.04 \exp(-x_1 - x_2) \times \exp(-0.045z_1^2 + 0.313z_1z_2 - 0.045z_2^2) \quad (17)$$

where $z_i = \Phi^{-1}(1 - \exp(-x_i))$, $i = 1, 2$. Whereas the PDF based on the Morgenstern model is a smooth surface bounded in the first quadrant of the sample space, the PDF based on the Nataf model exhibits strong warping along the edges $x_1 = 0$ and $x_2 = 0$ and it is unbounded at the origin. This behaviour is due to the high nonnormality of the exponential distribution at the origin. Although this behaviour may not be desirable in modelling physical phenomena, the Nataf model has the advantage that it can describe a much wider range of the correlation coefficient than the Morganstern model. For the present example, the admissible range of the correlation coefficient for the Nataf model is -0.645 to 1.000 , as indicated in Table 9.

DISCUSSION AND SUMMARY

As stated in the introduction, in many engineering applications the available information on the random variables is limited to the set of marginal distributions and the covariances. The multivariate distribution models presented in this paper should prove to be useful in describing the dependence between random variables in such applications. Both models presented are applicable to arbitrary number of random variables and, using the formulae developed in this paper, are easily evaluated in terms of the known marginals and correlation coefficients. This feature makes the proposed models particularly attractive for engineering analysis. The Morgenstern model provides a well behaving PDF, but is only applicable to variables with low correlation (i.e., within the range ± 0.3). The Nataf model is capable of describing a wider range of correlation coefficients, but the PDF may exhibit undesirable behaviour if the variables are highly nonnormal. These limitations and conditions for validity of the two models are further addressed in the paper.

One area in engineering where the Nataf model proves to be particularly attractive is in the evaluation of structural reliability. It is found convenient in such evaluations to transform the set of random variables X into the standard normal space⁷. This allows simple

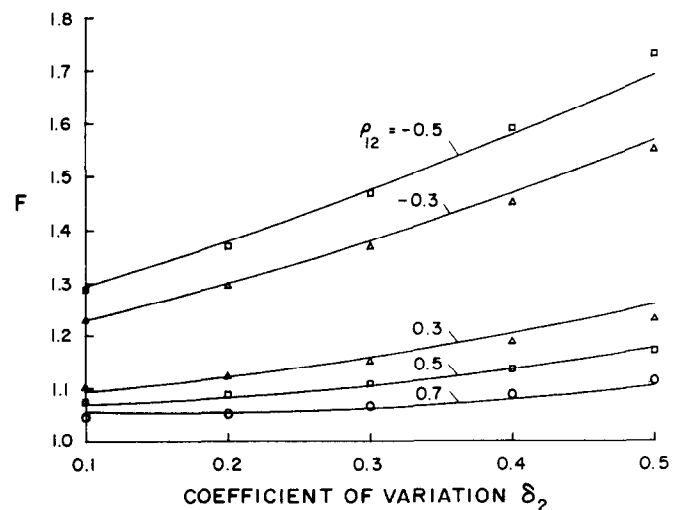
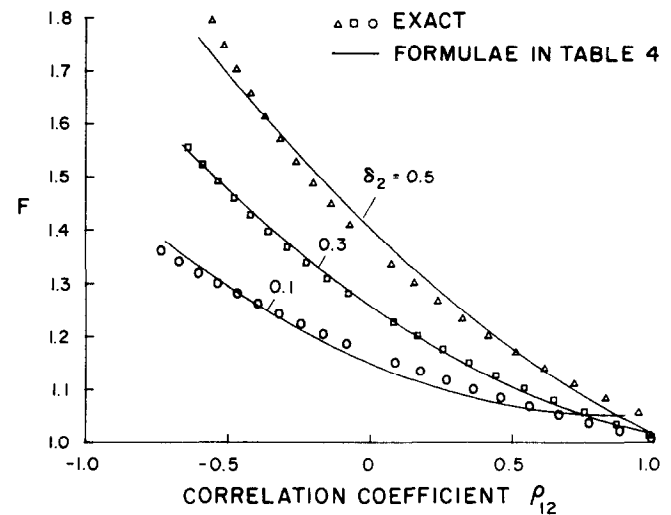

Fig. 3. F for X_1 shifted exponential and X_2 Type-II largest value

Table 9. Lower/upper bounds on ρ_{ij} for Nataf model

Marginal distribution	N	U	SE	SR	T1L	T1S
N	-1.000/1.000	-0.977/0.977	-0.903/0.903	-0.986/0.986	-0.969/0.969	-0.969/0.969
U	-0.977/0.977	-0.999/0.999	-0.886/0.886	-0.970/0.970	-0.936/0.936	-0.936/0.936
SE	-0.903/0.903	-0.866/0.866	-0.645/1.000	-0.819/0.957	-0.780/0.981	-0.981/0.780
SR	-0.986/0.986	-0.970/0.970	-0.819/0.957	-0.947/1.000	-0.915/0.993	-0.993/0.915
T1L	-0.969/0.969	-0.936/0.936	-0.780/0.981	-0.915/0.993	-0.886/1.000	-1.000/0.886
T1S	-0.969/0.969	-0.936/0.936	-0.981/0.780	-0.983/0.915	-1.000/0.886	-0.886/1.000

approximations of the reliability, even for very large number of random variables. Using the Nataf model, this transformation is easily accomplished by the marginal transformations of equation (10) and by a linear transformation of the variables Z . By contrast, the Morgenstern distribution would require a rather complicated transformation involving the conditional distributions⁷. Application of the Nataf model to the structural reliability problem is described in Refs 6 and 8.

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APPENDIX A

To prove $0 < Q < 0.5$, it is convenient to introduce the standard variate $U = (X - \mu)/\sigma$ with CDF $F_U(u) =$

$F_X(\sigma u + \mu)$. Equation (4) then can be written

$$\begin{aligned}
 Q &= \int_{-\infty}^{\infty} u f_U(u) F_U(u) du \\
 &= \int_{-\infty}^0 u f_U(u) F_U(u) du + \int_0^{\infty} u f_U(u) F_U(u) du \\
 &> F_U(0) \int_{-\infty}^0 u f_U(u) du + F_U(0) \int_0^{\infty} u f_U(u) du \\
 &= F_U(0) E[U] = 0
 \end{aligned} \quad (18)$$

which proves the lower bound. The upper bound is obtained by writing

$$\begin{aligned}
 Q &= \int_{-\infty}^0 u f_U(u) F_U(u) du + \int_0^{\infty} u f_U(u) F_U(u) du \\
 &< \int_0^{\infty} u f_U(u) du \\
 &= \int_0^{\infty} |u| f_U(u) du \\
 &= \frac{1}{2} E[|U|] < \frac{1}{2} \sqrt{E[U^2]} = 0.5
 \end{aligned} \quad (19)$$

If U_1 and U_2 denote two independent standard variates with identical distributions $F_U(u)$, one interesting result from the preceding analysis is

$$\begin{aligned}
 Q &= \int_{-\infty}^{\infty} u f_U(u) F_U(u) du = \frac{1}{2} \int_{-\infty}^{\infty} u dF_U(u)^2 \\
 &= \frac{1}{2} E[\max(U_1, U_2)]
 \end{aligned} \quad (20)$$

which implies $0 < E[\max(U_1, U_2)] < 1$ for any two such variables.

APPENDIX B

The proofs of Lemmas 1-7 are as follows:

Proof of Lemma 1. It is sufficient to show that the derivative $d\rho_{ij}/d\rho'_{ij}$ is positive. Omitting subscripts on ρ_{ij} and ρ'_{ij} for clarity,

$$\begin{aligned}
 \frac{d\rho}{d\rho'} &= \frac{1}{\sigma_i \sigma_j} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} x_i x_j \frac{\partial \phi_2(z_i, z_j, \rho')}{\partial \rho'} dz_i dz_j \\
 &= \frac{1}{\sigma_i \sigma_j} \int_{-\infty}^{\infty} x_i \int_{-\infty}^{\infty} x_j \frac{\partial^2 \phi_2(z_i, z_j, \rho')}{\partial z_i \partial z_j} dz_j dz_i
 \end{aligned}$$

$$\begin{aligned}
 &= \frac{1}{\sigma_i \sigma_j} \int_{-\infty}^{\infty} x_i \left\{ \left[x_j \frac{\partial \varphi_2(z_i, z_j, \rho')}{\partial z_i} \right]_{-\infty}^{\infty} \right. \\
 &\quad \left. - \int_{-\infty}^{\infty} \frac{dx_j}{dz_j} \frac{\partial \varphi_2(z_i, z_j, \rho')}{\partial z_i} dz_j \right\} dz_i \\
 &= -\frac{1}{\sigma_i \sigma_j} \int_{-\infty}^{\infty} \frac{dx_j}{dz_j} \int_{-\infty}^{\infty} x_i \frac{\partial \varphi_2(z_i, z_j, \rho')}{\partial z_i} dz_i dz_j \\
 &= -\frac{1}{\sigma_i \sigma_j} \int_{-\infty}^{\infty} \frac{dx_j}{dz_j} \left\{ \left[x_i \varphi_2(z_i, z_j, \rho') \right]_{-\infty}^{\infty} \right. \\
 &\quad \left. - \int_{-\infty}^{\infty} \frac{dx_i}{dz_i} \varphi_2(z_i, z_j, \rho') dz_i \right\} dz_j \\
 &= \frac{1}{\sigma_i \sigma_j} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \frac{dx_i}{dz_i} \frac{dx_j}{dz_j} \varphi_2(z_i, z_j, \rho') dz_i dz_j \quad (21)
 \end{aligned}$$

In this derivation, use is made of the fact that

$$\left[x_j \frac{\partial \varphi_2(z_i, z_j, \rho')}{\partial z_i} \right]_{-\infty}^{\infty} \quad \text{and} \quad [x_i \varphi_2(z_i, z_j, \rho')]_{-\infty}^{\infty}$$

are identically zero. This is obvious for marginal distributions with bounded tails. For distributions with unbounded tails, $f_{X_i}(x_i)$ must decay faster than x_i^{-3} to have a finite variance. Using this fact, examination of the asymptotic behaviour of the products

$$x_j \frac{\partial \varphi_2(z_i, z_j, \rho')}{\partial z_i} \quad \text{and} \quad x_i \varphi_2(z_i, z_j, \rho')$$

shows that both these products approach zero as z_i and z_j approach infinity. From equation (10), since x_i is a strictly increasing function of z_i , it follows that dx_i/dz_i , dx_j/dz_j and, therefore, the integral are all positive, which proves the lemma.

Proof of Lemma 2. This follows from simple substitution in equation (12). This property implies that X_i and X_j are considered independent if they are uncorrelated. Together with Lemma 1, this also shows that the algebraic sign of ρ'_{ij} is the same as that of ρ_{ij} .

The proof of Lemma 3 can be found in Ref. 9.

Proof of Lemma 4. For normal X_i , $Z_i = (X_i - \mu_i)/\sigma_i$ and equation (12) reduces to

$$\begin{aligned}
 \rho_{ij} &= \int_{-\infty}^{\infty} \left(\frac{x_j - \mu_j}{\sigma_j} \right) \varphi(z_j) \int_{-\infty}^{\infty} z_i \frac{\varphi_2(z_i, z_j, \rho'_{ij})}{\varphi(z_j)} dz_i dz_j \\
 &= \int_{-\infty}^{\infty} \left(\frac{x_j - \mu_j}{\sigma_j} \right) \varphi(z_j) \rho'_{ij} z_j dz_j \\
 &= \frac{1}{\sigma_j} E[X_j Z_j] \rho'_{ij} \quad (22)
 \end{aligned}$$

which shows that F is a function of the distribution of X_j only.

Proof of Lemma 5. Let $\bar{X}_k = a_k + b_k X_k$ and $\bar{Z}_k = \Phi^{-1}[F_{\bar{X}_k}(\bar{X}_k)]$, $k = i, j$, where a_k and b_k are arbitrary constants and $b_k > 0$. Using superposed bars to denote the properties of $\bar{X}_k$, $\bar{\mu}_k = a_k + b_k \mu_k$ and $\bar{\sigma}_k = b_k \sigma_k$. It is easy to see that $(\bar{x}_k - \bar{\mu}_k)/\bar{\sigma}_k = (x_k - \mu_k)/\sigma_k$. Also, since $F_{\bar{X}_k}(\bar{x}_k) = F_{X_k}(x_k)$, then $\bar{z}_k = z_k$. Now, equation (12) for X_i and X_j can be written

$$\bar{\rho}_{ij} = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \left(\frac{x_i - \mu_i}{\sigma_i} \right) \left(\frac{x_j - \mu_j}{\sigma_j} \right) \varphi_2(z_i, z_j, \bar{\rho}'_{ij}) dz_i dz_j \quad (23)$$

Since $\bar{\rho}_{ij} = \rho_{ij}$, it follows that $\bar{\rho}'_{ij} = \rho'_{ij}$, which proves that F is invariant.

Proof of Lemma 8. This is a direct result of Lemma 5, since for a Group 1 distribution the variable can be linearly transformed to a standard form.

Proof of Lemma 7. This follows from Lemma 5, since for a Group 2 distribution the variable can be scaled to have a unit mean, in which case the shape of the distribution is completely described by the coefficient of variation.